DATA LOADING IN SNOWFLAKE

Internal stages:

a. User stage

Whenever any user gets created in snowflake, it internally creates by default a user stage

Let's see in practical by loading a file into user stage

Check the file 'subjects' what data is present (you can take your own file)

	A	B	C	D
1	'Amit'	Physics'	'Maths'	'Biology'
2	'Sharan'	'English'	'Economics'	'History'
3	'James'	Statistics'	'Psychology	'Biology'
4	'Ashley'	'English'	'Mathemati	'Statistics'
5				
6				

Now run the below in your snowsql for putting file into user stage (note: it varies from stage to stage)

Put file ://< filepath> **@~**/<user> auto_compress=false;

```
SAIKPI#COMPUTE_WH@SAI_K.PUBLIC>put file://C:\Users\GudiboinaSaiKishore\Desktop\subjects.csv @~/SAI_K auto_compress=false;
+-------------+-------------+------------+------------+-------------------+-------------------+----------+---------+
| source      | target      | source_size | target_size | source_compression | target_compression | status   | message |
|-------------+-------------+------------+------------+-------------------+-------------------+----------+---------|
| subjects.csv | subjects.csv |        164 |        176 | NONE              | NONE              | UPLOADED |         |
+-------------+-------------+------------+------------+-------------------+-------------------+----------+---------+
1 Row(s) produced. Time Elapsed: 3.311s
```

Checking list of user stages

```
SAIKPI#COMPUTE_WH@SAI_K.PUBLIC>list @~/SAI_K;
+--------------------+------+----------------------------------+-------------------------------+
| name               | size | md5                              | last_modified                 |
|--------------------+------+----------------------------------+-------------------------------|
| SAI_K/subjects.csv |  176 | f345dc8f93c23d3c9f8b098cb59aceae | Sun, 5 May 2024 14:30:10 GMT |
+--------------------+------+----------------------------------+-------------------------------+
1 Row(s) produced. Time Elapsed: 0.205s
```

In case of user stage, you have to explicitly give column names to check instead of * as shown

```
SAIKPI#COMPUTE_WH@SAI_K.PUBLIC>select t.$1, t.$2, t.$3, t.$4 from @~/SAI_K t;
+----------+-------------+---------------+-------------+
| $1       | $2          | $3            | $4          |
|----------+-------------+---------------+-------------|
| 'Amit'   | Physics'    | 'Maths'       | 'Biology'   |
| 'Sharan' | 'English'   | 'Economics'   | 'History'   |
| 'James'  | Statistics' | 'Psychology'  | 'Biology'   |
| 'Ashley' | 'English'   | 'Mathematics' | 'Statistics' |
+----------+-------------+---------------+-------------+
4 Row(s) produced. Time Elapsed: 0.804s
```

Now moving data from user stage to any table

Firstly creating a table

```
SAIKPI#COMPUTE_WH@SAI_K.PUBLIC>create table subjects (name string, sub1 string, sub2 string, sub3 string) ;
+-------------------------------------+
| status                              |
|-------------------------------------|
| Table SUBJECTS successfully created. |
+-------------------------------------+
1 Row(s) produced. Time Elapsed: 1.939s
SAIKPI#COMPUTE_WH@SAI_K.PUBLIC>
```

```
SAIKPI#COMPUTE_WH@SAI_K.PUBLIC>select * from subjects ;
+------+------+------+------+
| NAME | SUB1 | SUB2 | SUB3 |
|------+------+------+------|
+------+------+------+------+
0 Row(s) produced. Time Elapsed: 0.132s
SAIKPI#COMPUTE_WH@SAI_K.PUBLIC>
```

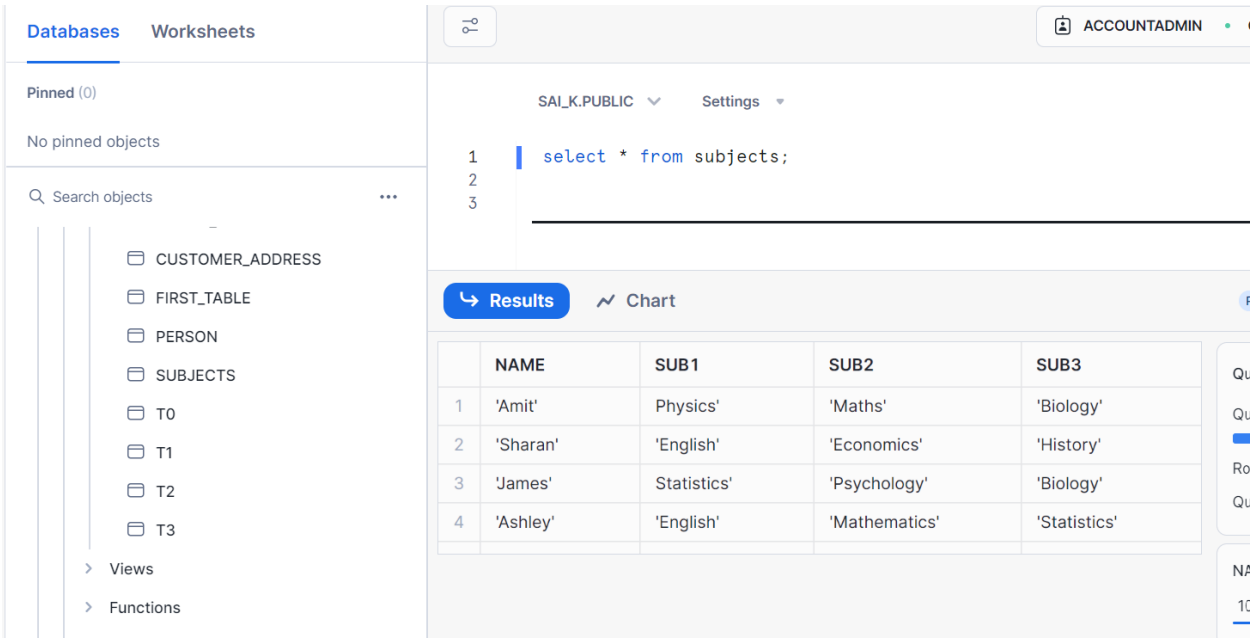
Copying the data from user stage & can give file format option here, if file type is not mentioned then default is csv

```
SAIKPI#COMPUTE_WH@SAI_K.PUBLIC>copy into subjects from @~/SAI_K;
+-------------------+--------+------------+------------+-------------+------------+------------+-----------------+------------
---------+------------------------+
| file              | status | rows_parsed | rows_loaded | error_limit | errors_seen | first_error | first_error_line | first_error_
character | first_error_column_name |
|-------------------+--------+------------+------------+-------------+------------+------------+-----------------+------------
---------+------------------------|
| SAI_K/subjects.csv | LOADED |           4 |           4 |           1 |           0 | NULL       |             NULL |
     NULL | NULL                   |
+-------------------+--------+------------+------------+-------------+------------+------------+-----------------+------------
---------+------------------------+
1 Row(s) produced. Time Elapsed: 1.165s
SAIKPI#COMPUTE_WH@SAI_K.PUBLIC>
```

```
SAIKPI#COMPUTE_WH@SAI_K.PUBLIC>select * from subjects ;
+----------+-------------+---------------+-------------+
| NAME     | SUB1        | SUB2          | SUB3        |
|----------+-------------+---------------+-------------|
| 'Amit'   | Physics'    | 'Maths'       | 'Biology'   |
| 'Sharan' | 'English'   | 'Economics'   | 'History'   |
| 'James'  | Statistics' | 'Psychology'  | 'Biology'   |
| 'Ashley' | 'English'   | 'Mathematics' | 'Statistics' |
+----------+-------------+---------------+-------------+
4 Row(s) produced. Time Elapsed: 0.242s
```

Let’s check in snowflake

I



Let’s clean up

```
SAIKPI#COMPUTE_WH@SAI_K.PUBLIC>drop table subjects ;
+-------------------------------+
| status                        |
|-------------------------------|
| SUBJECTS successfully dropped. |
+-------------------------------+
1 Row(s) produced. Time Elapsed: 0.253s
SAIKPI#COMPUTE_WH@SAI_K.PUBLIC>
```

```
SAIKPI#COMPUTE_WH@SAI_K.PUBLIC>remove @~/SAI_K pattern='.*.csv' ;
+--------------------+---------+
| name               | result  |
|--------------------+---------|
| SAI_K/subjects.csv | removed |
+--------------------+---------+
1 Row(s) produced. Time Elapsed: 0.252s
```

```
SAIKPI#COMPUTE_WH@SAI_K.PUBLIC>list @~/SAI_K;
+------+------+-----+---------------+
| name | size | md5 | last_modified |
|------+------+-----+---------------|
+------+------+-----+---------------+
0 Row(s) produced. Time Elapsed: 0.204s
SAIKPI#COMPUTE_WH@SAI_K.PUBLIC>
```

b. Table stage

Table stage is the stage created default by snowflake when a table is created

Let's check in practical

Let's check the data in the file salary (select your file)

	A	B	C
1	20	10000	
2	30	40000	
3	35	20000	
4	28	30000	
5			
6			
7			
8			

Let's create a table salary

```
SAIKPI#COMPUTE_WH@SAI_K.PUBLIC>create table salary (Age number , Sal number) ;
+-----------------------------------+
| status                            |
|-----------------------------------|
| Table SALARY successfully created. |
+-----------------------------------+
1 Row(s) produced. Time Elapsed: 1.198s
SAIKPI#COMPUTE_WH@SAI_K.PUBLIC>
```

Note:

Via snowsql, table stage should have same name as table, cannot be altered or dropped, not a DB object

Now put the file into table stage

```
SAIKPI#COMPUTE_WH@SAI_K.PUBLIC>put file://C:\Users\GudiboinaSaiKishore\Desktop\salary.csv @%salary auto_compress=false;
+------------+------------+-------------+-------------+--------------------+--------------------+----------+---------+
| source     | target     | source_size | target_size | source_compression | target_compression | status   | message |
|------------+------------+-------------+-------------+--------------------+--------------------+----------+---------|
| salary.csv | salary.csv |          36 |          48 | NONE               | NONE               | UPLOADED |         |
+------------+------------+-------------+-------------+--------------------+--------------------+----------+---------+
1 Row(s) produced. Time Elapsed: 2.001s
SAIKPI#COMPUTE_WH@SAI_K.PUBLIC>
```

Verifying table

```
SAIKPI#COMPUTE_WH@SAI_K.PUBLIC>select * from salary ;
+-----+-----+
| AGE | SAL |
|-----+-----|
+-----+-----+
0 Row(s) produced. Time Elapsed: 0.346s
SAIKPI#COMPUTE_WH@SAI_K.PUBLIC>
```

List the files in table stages (this implies pointing to a location)

```
SAIKPI#COMPUTE_WH@SAI_K.PUBLIC>list @%salary;
+------------+------+----------------------------------+-----------------------------+
| name       | size | md5                              | last_modified               |
|------------+------+----------------------------------+-----------------------------|
| salary.csv |   48 | b35c7e6eba268b2fc015c14d905a6eb1 | Sun, 5 May 2024 15:57:49 GMT |
+------------+------+----------------------------------+-----------------------------+
1 Row(s) produced. Time Elapsed: 0.306s
SAIKPI#COMPUTE_WH@SAI_K.PUBLIC>
```

```
SAIKPI#COMPUTE_WH@SAI_K.PUBLIC>select t.$1, t.$2 from @%salary t;
+----+-------+
| $1 | $2    |
|----+-------|
| 21 | 20000 |
| 31 | 50000 |
| 32 | 25000 |
| 25 | 29000 |
+----+-------+
4 Row(s) produced. Time Elapsed: 0.711s
SAIKPI#COMPUTE_WH@SAI_K.PUBLIC>
```

Direct select * works in table stages

```
SAIKPI#COMPUTE_WH@SAI_K.PUBLIC>select * from @%salary t;
+-----+-------+
| AGE | SAL   |
|-----+-------|
| 21  | 20000 |
| 31  | 50000 |
| 32  | 25000 |
| 25  | 29000 |
+-----+-------+
4 Row(s) produced. Time Elapsed: 0.443s
SAIKPI#COMPUTE_WH@SAI_K.PUBLIC>
```

Use copy command to load the data in to table_stage

Can specify file format option here. If file type is not given, default is csv

```
SAIKPI#COMPUTE_WH@SAI_K.PUBLIC>copy into salary from @%salary ;
+------------+--------+-------------+-------------+-------------+-------------+-------------+------------------+--------------------
--+------------------------+
| file       | status | rows_parsed | rows_loaded | error_limit | errors_seen | first_error | first_error_line | first_error_characte
r | first_error_column_name |
|------------+--------+-------------+-------------+-------------+-------------+-------------+------------------+--------------------
--+------------------------|
| salary.csv | LOADED |           4 |           4 |           1 |           0 | NULL        |             NULL |                  NUL
L | NULL                   |
+------------+--------+-------------+-------------+-------------+-------------+-------------+------------------+--------------------
--+------------------------+
1 Row(s) produced. Time Elapsed: 0.665s
SAIKPI#COMPUTE_WH@SAI_K.PUBLIC>
```

Validate

```
SAIKPI#COMPUTE_WH@SAI_K.PUBLIC>select * from salary ;
+-----+-------+
| AGE |   SAL |
|-----+-------|
|  21 | 20000 |
|  31 | 50000 |
|  32 | 25000 |
|  25 | 29000 |
+-----+-------+
4 Row(s) produced. Time Elapsed: 0.641s
SAIKPI#COMPUTE_WH@SAI_K.PUBLIC>
```

Let’s check in snowflake

SAI_K.PUBLIC Settings

```
1  select * from salary;
2
3
```

Results Chart PREVIEW

	AGE	SAL
1	21	20000
2	31	50000
3	32	25000
4	25	29000

Query De
Query du
Rows
Query ID
AGE

Cleaning up

Dropping table will remove the data from stage table_stage

```
SAIKPI#COMPUTE_WH@SAI_K.PUBLIC>drop table salary ;
+-----------------------------+
| status                      |
|-----------------------------|
| SALARY successfully dropped. |
+-----------------------------+
1 Row(s) produced. Time Elapsed: 0.205s
SAIKPI#COMPUTE_WH@SAI_K.PUBLIC>
```

OR To reload, we can remove the files by using syntax

Remove @%salary pattern='.*.csv’;

Check the stage data now

```
SAIKPI#COMPUTE_WH@SAI_K.PUBLIC>select * from @%salary t;
002001 (02000): SQL compilation error:
Object 'SALARY' does not exist or not authorized.
```

Named stages

Internal named stage is a database object and as a user we can create it

Let's see hands on

Let's check the file 'weather'

	A	B	C
1	warangal	80	
2	hyderabad	75	
3	raidurg	78	
4	hitech city	85	
5	kondapur	95	
6			
7			
8			

As for user stage and table stage we never created manually as they by default

But here, we had to create manually

```
SAIKPI#COMPUTE_WH@SAI_K.PUBLIC>create stage shared_stage ;
+-----------------------------------------------+
| status                                        |
|-----------------------------------------------|
| Stage area SHARED_STAGE successfully created. |
+-----------------------------------------------+
1 Row(s) produced. Time Elapsed: 0.221s
SAIKPI#COMPUTE_WH@SAI_K.PUBLIC>
```

```
SAIKPI#COMPUTE_WH@SAI_K.PUBLIC>show stages ;
+-------------------------------+--------------+--------------+-------------+-----+-----------------+-----------------
--+--------------+---------+--------+----------+-------+---------------------+---------------------+----------+--------
---------+-------------------+
| created_on                    | name         | database_name | schema_name | url | has_credentials | has_encryption_ke
y | owner        | comment | region | type     | cloud | notification_channel | storage_integration | endpoint | owner_r
ole_type | directory_enabled |
|-------------------------------+--------------+--------------+-------------+-----+-----------------+-----------------
--+--------------+---------+--------+----------+-------+---------------------+---------------------+----------+--------
---------+-------------------|
| 2024-05-06 01:26:11.372 -0700 | SHARED_STAGE | SAI_K        | PUBLIC      |     | N               | N
  | ACCOUNTADMIN |         | NULL   | INTERNAL | NULL  | NULL                | NULL                | NULL     | ROLE
        | N                 |
+-------------------------------+--------------+--------------+-------------+-----+-----------------+-----------------
--+--------------+---------+--------+----------+-------+---------------------+---------------------+----------+--------
---------+-------------------+
1 Row(s) produced. Time Elapsed: 0.116s
SAIKPI#COMPUTE_WH@SAI_K.PUBLIC>
```

SAI_K.PUBLIC Settings

```
1  select * from salary;
2
3  show stages;
```

Results Chart

	created_on	name	database_name	schema_n
1	2024-05-06 01:26:11.372 -070(	SHARED_STAGE	SAI_K	PUBLIC

Describe internal stage

```
SAIKPI#COMPUTE_WH@SAI_K.PUBLIC>desc stage shared_stage  ;
+--------------------+-------------------------------+---------------+-----------------+-------------------+
| parent_property    | property                      | property_type | property_value  | property_default |
|--------------------+-------------------------------+---------------+-----------------+-------------------|
| STAGE_FILE_FORMAT  | TYPE                          | String        | CSV             | CSV               |
| STAGE_FILE_FORMAT  | RECORD_DELIMITER              | String        | \n              | \n                |
| STAGE_FILE_FORMAT  | FIELD_DELIMITER               | String        | ,               | ,                 |
| STAGE_FILE_FORMAT  | FILE_EXTENSION                | String        |                 |                   |
| STAGE_FILE_FORMAT  | SKIP_HEADER                   | Integer       | 0               | 0                 |
| STAGE_FILE_FORMAT  | PARSE_HEADER                  | Boolean       | false           | false             |
| STAGE_FILE_FORMAT  | DATE_FORMAT                   | String        | AUTO            | AUTO              |
| STAGE_FILE_FORMAT  | TIME_FORMAT                   | String        | AUTO            | AUTO              |
| STAGE_FILE_FORMAT  | TIMESTAMP_FORMAT              | String        | AUTO            | AUTO              |
| STAGE_FILE_FORMAT  | BINARY_FORMAT                 | String        | HEX             | HEX               |
| STAGE_FILE_FORMAT  | ESCAPE                        | String        | NONE            | NONE              |
| STAGE_FILE_FORMAT  | ESCAPE_UNENCLOSED_FIELD       | String        | \\              | \\                |
| STAGE_FILE_FORMAT  | TRIM_SPACE                    | Boolean       | false           | false             |
| STAGE_FILE_FORMAT  | FIELD_OPTIONALLY_ENCLOSED_BY  | String        | NONE            | NONE              |
| STAGE_FILE_FORMAT  | NULL_IF                       | List          | [\\N]           | [\\N]             |
| STAGE_FILE_FORMAT  | COMPRESSION                   | String        | AUTO            | AUTO              |
| STAGE_FILE_FORMAT  | ERROR_ON_COLUMN_COUNT_MISMATCH | Boolean       | true            | true              |
| STAGE_FILE_FORMAT  | VALIDATE_UTF8                 | Boolean       | true            | true              |
| STAGE_FILE_FORMAT  | SKIP_BLANK_LINES              | Boolean       | false           | false             |
| STAGE_FILE_FORMAT  | REPLACE_INVALID_CHARACTERS    | Boolean       | false           | false             |
| STAGE_FILE_FORMAT  | EMPTY_FIELD_AS_NULL           | Boolean       | true            | true              |
```

Alter internal stage (good use case to load files via multiple users to multiple tables)

```
SAIKPI#COMPUTE_WH@SAI_K.PUBLIC>alter stage shared_stage rename to shared_stage2;
+-----------------------------------+
| status                            |
|-----------------------------------|
| Statement executed successfully.  |
+-----------------------------------+
1 Row(s) produced. Time Elapsed: 0.141s
SAIKPI#COMPUTE_WH@SAI_K.PUBLIC>
```

```
desc stage shared_stage2 ;
```

Results | Chart

	parent_property	property	property_type	property_value
1	STAGE_FILE_FORMAT	TYPE	String	CSV
2	STAGE_FILE_FORMAT	RECORD_DELIMITER	String	\n
3	STAGE_FILE_FORMAT	FIELD_DELIMITER	String	,
4	STAGE_FILE_FORMAT	FILE_EXTENSION	String	
5	STAGE_FILE_FORMAT	SKIP_HEADER	Integer	0
6	STAGE_FILE_FORMAT	PARSE_HEADER	Boolean	false

Let's put the file into named stage 'shared_stage2'

```
SAIKPI#COMPUTE_WH@SAI_K.PUBLIC>put file://C:\Users\GudiboinaSaiKishore\Desktop\snowflake\snowflake_udemy\dataloading\we
                               ather.csv @shared_stage2 auto_compress=false;
+-------------+-------------+-------------+-------------+--------------------+--------------------+----------+---------+
| source      | target      | source_size | target_size | source_compression | target_compression | status   | message |
|-------------+-------------+-------------+-------------+--------------------+--------------------+----------+---------|
| weather.csv | weather.csv |          68 |          80 | NONE               | NONE               | UPLOADED |         |
+-------------+-------------+-------------+-------------+--------------------+--------------------+----------+---------+
1 Row(s) produced. Time Elapsed: 3.837s
SAIKPI#COMPUTE_WH@SAI_K.PUBLIC>
```

Select columns from internal named stage

```
SAIKPI#COMPUTE_WH@SAI_K.PUBLIC>select t.$1, t.$2  from @shared_stage2 t;
+-------------+----+
| $1          | $2 |
|-------------+----|
| warangal    | 80 |
| hyderabad   | 75 |
| raidurg     | 78 |
| hitech city | 85 |
| kondapur    | 95 |
+-------------+----+
5 Row(s) produced. Time Elapsed: 0.849s
SAIKPI#COMPUTE_WH@SAI_K.PUBLIC>
```

Select * doesn't work here, it only works for table stage

```
SAIKPI#COMPUTE_WH@SAI_K.PUBLIC>select * from @shared_stage2 t;
001080 (42601): SQL compilation error: error line 1 at position 0
SELECT with no columns
SAIKPI#COMPUTE_WH@SAI_K.PUBLIC>
```

Now, let's create a table to load internal named stage data

```
SAIKPI#COMPUTE_WH@SAI_K.PUBLIC>create table weather (city string, temperature number);
+-------------------------------------+
| status                              |
|-------------------------------------|
| Table WEATHER successfully created. |
+-------------------------------------+
1 Row(s) produced. Time Elapsed: 1.422s
SAIKPI#COMPUTE_WH@SAI_K.PUBLIC>
```

Copy the data from internal named stage to table

```
SAIKPI#COMPUTE_WH@SAI_K.PUBLIC>copy into weather from @shared_stage2 ;
+--------------------------+--------+-------------+-------------+-------------+-------------+-------------+------------
------+----------------------+-----------------------+
| file                     | status | rows_parsed | rows_loaded | error_limit | errors_seen | first_error | first_error
_line | first_error_character | first_error_column_name |
|--------------------------+--------+-------------+-------------+-------------+-------------+-------------+------------
------+----------------------+-----------------------|
| shared_stage2/weather.csv | LOADED |           5 |           5 |           1 |           0 | NULL        |
 NULL |                 NULL | NULL                  |
+--------------------------+--------+-------------+-------------+-------------+-------------+-------------+------------
------+----------------------+-----------------------+
1 Row(s) produced. Time Elapsed: 0.629s
SAIKPI#COMPUTE_WH@SAI_K.PUBLIC>
```

```
SAIKPI#COMPUTE_WH@SAI_K.PUBLIC>select * from weather ;
+-------------+-------------+
| CITY        | TEMPERATURE |
|-------------+-------------|
| warangal    |          80 |
| hyderabad   |          75 |
| raidurg     |          78 |
| hitech city |          85 |
| kondapur    |          95 |
+-------------+-------------+
5 Row(s) produced. Time Elapsed: 0.250s
SAIKPI#COMPUTE_WH@SAI_K.PUBLIC>
```

```
select * from weather ;
```

Results | Chart

	CITY	TEMPERATURE
1	warangal	80
2	hyderabad	75
3	raidurg	78
4	hitech city	85
5	kondapur	95

Cleaning up

```
SAIKPI#COMPUTE_WH@SAI_K.PUBLIC>drop stage shared_stage2 ;
+-------------------------------------+
| status                              |
|-------------------------------------|
| SHARED_STAGE2 successfully dropped. |
+-------------------------------------+
1 Row(s) produced. Time Elapsed: 0.126s
SAIKPI#COMPUTE_WH@SAI_K.PUBLIC>
```

```
SAIKPI#COMPUTE_WH@SAI_K.PUBLIC>drop table weather ;
+-------------------------------+
| status                        |
|-------------------------------|
| WEATHER successfully dropped. |
+-------------------------------+
1 Row(s) produced. Time Elapsed: 0.173s
SAIKPI#COMPUTE_WH@SAI_K.PUBLIC>
```

```
select * from weather ;
```

Results | Chart

002003 (42S02): SQL compilation error:

Object 'WEATHER' does not exist or not authorized.

Used cases of internal stages:

INTERNAL STAGES

User Stage	Table Stage	Named Stage
Use Case: If a user wants to load the data from a local file to multiple tables	Use Case: Specific to table, if the file has to be loaded to a single table.	Use Case: If multiple users wants to load files to multiple tables

External stages

External stages deal with the files which are **located externally at the cloud location** where as internal stages deals with the local files in the system.

Let's see hands on

Creating an external stage by giving name as 'contacts_stage' and URL is referring to S3 bucket

```
CREATE OR REPLACE STAGE contacts_stage url='s3://snowflake-docs/tutorials/dataloading/' ;
```

Results | Chart | PREVIEW

	status
1	Stage area CONTACTS_STAGE successfully created.

Query Details

Query duration 1.0s

Rows 1

Query ID 01b424b6-3201-130c-0...

status A

100% filled

```
DESC STAGE contacts_stage;
```

Results | Chart | PREVIEW

	parent_property	property	property_type	property_value
1	STAGE_FILE_FORMAT	TYPE	String	CSV
2	STAGE_FILE_FORMAT	RECORD_DELIMITER	String	\n
3	STAGE_FILE_FORMAT	FIELD_DELIMITER	String	,
4	STAGE_FILE_FORMAT	FILE_EXTENSION	String	
5	STAGE_FILE_FORMAT	SKIP_HEADER	Integer	0
6	STAGE_FILE_FORMAT	PARSE_HEADER	Boolean	false

Query Details

Query duration 57ms

Rows 34

Query ID 01b424b9-3201-130c-0...

parent_property A

STAGE_FILE_FORMAT 23

```
SHOW stages ;
```

Results | Chart

	created_on	name	database_name	schem
1	2024-05-06 01:54:07.489 -0700	CONTACTS_STAGE	SAI_K	PUBLI

List of files in different formats in our stage

```
LIST @contacts_stage;
```

Results | Chart | PREVIEW

	name	size	md5
	s3://snowflake-docs/tutorials/dataloading/basicsampledata.zip	4850	95d063bl
2	s3://snowflake-docs/tutorials/dataloading/cities.parquet	866	88ad5cdf
3	s3://snowflake-docs/tutorials/dataloading/contacts.json	965	3cac26e4
4	s3://snowflake-docs/tutorials/dataloading/contacts1.csv	694	0d58433
5	s3://snowflake-docs/tutorials/dataloading/contacts2.csv	763	e05dd37c
6	s3://snowflake-docs/tutorials/dataloading/contacts3.csv	771	2ea4086f

Query Det

Query dur

Rows

Query ID

name

100% fille

As external stage is a database object, we can rename

```
ALTER STAGE contacts_stage rename to contacts_stage1;
```

Results | Chart | PREVIEW

	status
1	Statement executed successfully.

Query Details

Query duration

Rows

Query ID 01b424bd-3201-130a

status

100% filled

Creating a table

```
create or replace table CONTACTS (
  id integer,
  last_name string,
  first_name string,
  company string,
  email string,
  workphone string,
  cellphone string,
  streetaddress string,
  city string,
  postalcode string);
```

Results | Chart | PREVIEW

	status
1	Table CONTACTS successfully created.

Query Details

Query duration

Rows

Query ID

Copying the data from external stage to table

Note: we are copying data of contacts5 csv data which has delimiter |, type csv

```
COPY INTO CONTACTS
  FROM @contacts_stage1
  file_format = ( type = 'CSV'
          field_delimiter = '|'
          skip_header = 1)
  pattern='.*.contacts5.csv';
```

Results | Chart | PREVIEW

	file	status	rows_parsed
1	s3://snowflake-docs/tutorials/dataloading/contacts5.csv	LOADED	6

Query Detai
Query durat
Rows
Query ID

```
select * from CONTACTS ;
```

Results | Chart | PREVIEW

	ID	LAST_NAME	FIRST_NAME	COMPANY	EMAIL
1	20	Michelle	Dickson	Ut Limited	Duis.digniss
2	21	Lance	Harper	Rutrum Lorem Limited	Sed.neque@
3	22	Keely	Pace	Eleifend Limited	ante.bibend
4	23	Sage	Leblanc	Egestas A Consulting	dapibus@el
5	24	Marny	Holt	Urna Nec Luctus Associates	ornare@vita
6	25	Holly	Park	Mauris PC	Vestibulum.a

Query Details
Query duration 262ms
Rows 6
Query ID 01b424c6-3201-130a-0...

ID

We got our data

To conclude, we brought the data into table from external source location (here we had taken s3 bucket folder which we had files in it)

Cleaning up

```
drop table CONTACTS ;
```

Results | Chart | PREVIEW

	status
1	CONTACTS successfully dropped.

Query Details

Query duration

Rows

Query ID 01b424c7-3201-130

```
drop stage contacts_stage1 ;
```

Results | Chart | PREVIEW

	status
1	CONTACTS_STAGE1 successfully dropped.

Query Details

Query duration

Rows